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CancerImmunoDB: A Comprehensive Resource for Immunomodulatory Effects of Anticancer Drugs

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SCHOLARONE MANUSCRIPTS

CancerImmunoDB: A Comprehensive Resource for Immunomodulatory Effects of Anticancer Drugs

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ABSTRACT

The response of a patient to anti-cancer therapy depends on the interaction between the tumor microenvironment and drugs, as well as the subsequent immune modulation. These parameters hold significant prognostic value and serve as important predictive factors for disease outcomes and survival. Thus, we developed **CancerImmunoDB** as an extensive resource for examining the immunomodulatory effects of anticancer drugs and antineoplastic agents. This database contains manually curated, experimentally verified data from published research articles, encompassing 1379 cell line studies, 420 animal studies, and 165 human studies. Each entry provides structured information on the immunomodulatory properties of anticancer drugs, including immune components whose function or number was altered following drug treatment, along with details such as assay type, drug dose, and experimental system. Additionally, the database includes drug structures predicted from SMILES format, enhancing its utility for molecular studies. Entries are cross-referenced with external databases like PubChem, DrugBank, and KEGG, enabling users to navigate seamlessly. **CancerImmunoDB** features a user-friendly web interface that supports basic and conditional searches, browsing, as well as molecular similarity and substructure search tools. This resource aims to facilitate the development of novel therapeutic strategies and advance the understanding of drug-immune system interactions. **CancerImmunoDB** is freely available at https://www.canimmdb.online/.

Keywords: Anticancer drugs, Immunomodulation, Drug-immune interaction, Database, Tumor microenvironments

1. INTRODUCTION

Cancer is a heterogeneous group of diseases characterized by uncontrolled growth and the spread of abnormal cells [1]. An estimated 20 million new cases of cancer were reported globally in 2022, accounting for 2.5 million deaths from lung cancer, 2.3 million deaths from female breast cancer, and 1.9 million deaths in colorectal cancer -making these the most prevalent cancer types [2]. According to the GLOBOCAN 2022 study, the incidence of new cancer cases could rise to 35 million by 2050, posing a major global health concern [3,4]. This projected increase will have profound social and economic implications, necessitating the implementation of tailored cancer control programs. Furthermore, regions such as Asia, Latin America, the Caribbean, and Africa face significant challenges in terms of prevention, treatment, and mitigation of its economic burden [5]. Limited access to screening programs, therapeutic strategies, and effective prevention methods exacerbates the global cancer burden [6].

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Over the past few decades, tumor microenvironment (TME) has garnered widespread attention in cancer research. The TME is a complex and dynamic environment composed of a heterogeneous repertoire of cell types including cancer cells, stromal cells, endothelial cells, and immune cells [7]. Among the various cell populations in the TME, immune cells significantly influence tumorigenesis by either promoting or inhibiting tumor growth. Effector immune cells, such as CD8+ cytotoxic T cells, CD4+ T cells, natural killer (NK) cells, dendritic cells (DCs), M1-polarized macrophages and N1-polarized neutrophils contribute antitumor immune responses [8]. In contrast, immunosuppressive cells such as type 2-polarized macrophages (M2), myeloid-derived suppressor cell (MDSC), and regulatory T cells regulatory T cells (Tregs), displays pro tumor activity [9,10]. Given the critical roles of tumor-associated immune cells in cancer pathogenesis, strategies targeting the TME represent promising therapeutic targets[11]. Scientific investigations over the past three decades into both targeted and conventional anticancer agents have revealed their potential immunomodulatory effects, encompassing both immunostimulatory and immunosuppressive properties [12]. These antineoplastic agents not only act directly on tumours but also interact with immune cells, altering their functions in ways that can have substantial therapeutic implications [13].

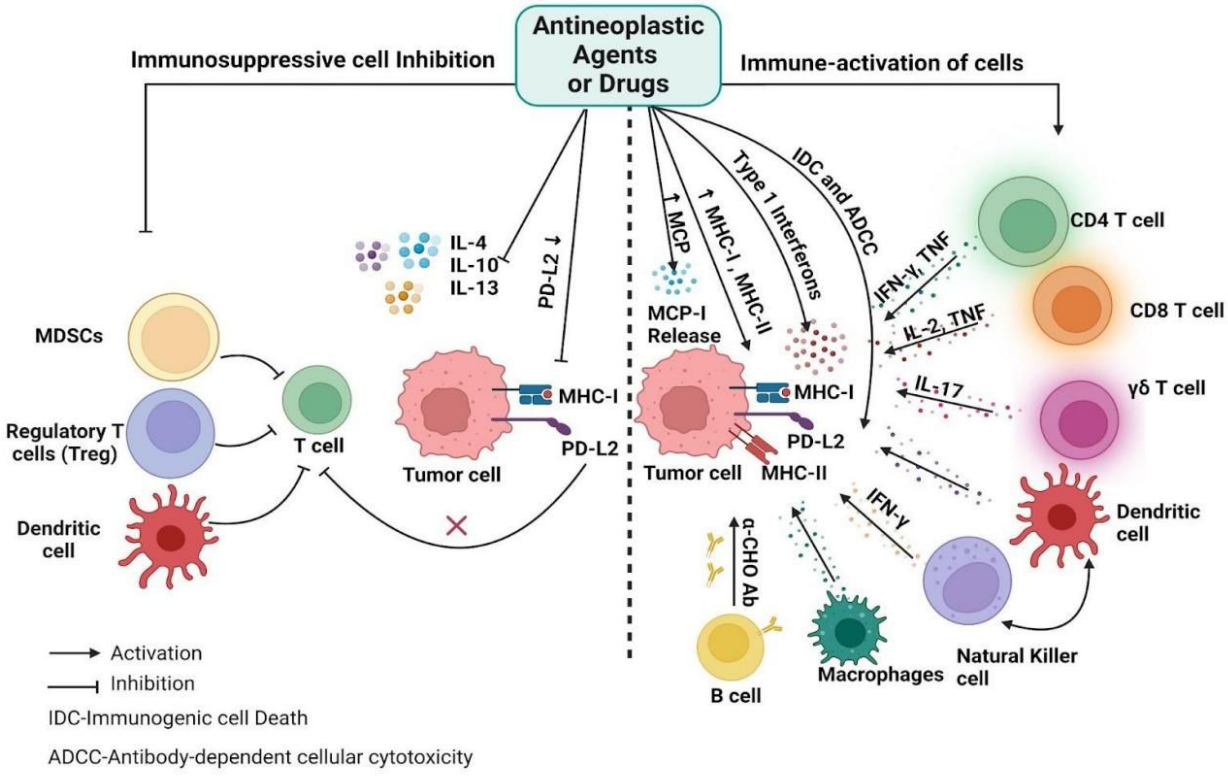


Fig 1. Mechanisms of Antineoplastic Agents in Modulating the Tumor Microenvironment (TME)

The mechanisms by which anticancer agents alter immune function include depleting immunosuppressive populations, both in circulation and within tumors, such as myeloid-derived suppressor cells (MDSCs) and regulatory T cells [14–16]. These agents also downregulate PD-L2 mediated immunosuppression [17], trigger immunogenic cell death [18], and induce antibody-dependent cellular cytotoxicity (ADCC) [19,20] . Additionally, they activate innate immune mechanisms, including the stimulation of of natural killer cells [21–26], dendritic cells [27] and macrophages [28]. (Figure 1)

Anticancer agents have been shown to induce tumour-specific immune responses such as IL-17–producing γδ T cells [29], IFN-γ producing CD8 T cells [30] and $CD4^+$ T helper 17 (Th17) cells [31]. Furthermore, they suppress protumor cytokines, including IL-4, IL-10, and IL-13 [32] while inducing type I interferon [33], class I MHC [34], class II MHC [35], MCP-1 [36,37]. Together, these experimental findings demonstrate that anticancer agents can significantly alter immune-related characteristics, such as immune cell populations, phenotypes, and overall function of the host immune system [38]. These immune associated parameters hold prognostic value and may serve as critical predictive factors for disease outcome and survival [39]. For example, in colorectal cancer, CD16+ myeloid cell infiltration is associated with improved overall survival, even after controlling for other prognostic factors [40]. Furthermore, in Hodgkin's lymphoma, cytotoxic cell infiltration serves as a prognostic marker correlated with a decreased overall survival rate [41]. Such immune associated parameters can be identified and clinically exploited for therapeutic interventions, paying the way for personalized medicine approaches.

Over the past few years, several databases have been developed to catalog information on cancer-immune interface, including Checkpoint Therapeutic Target Database (CKTTD) [42] CancerImmunityQTL [43], TCMIO: A Comprehensive Database of Traditional Chinese Medicine on Immuno-Oncology [44], and TIMEDB [45]. However, to the best of our knowledge, no dedicated web based computational resource exists that systematically catalogs the scattered literature on immunomodulation by anticancer drugs across different systems like cell line, animal models and patients.

To address this gap, we developed a manually curated resource to study tumor specific antitumor immune response of anticancer agents and drugs. This database provides structured information on immune components whose number or functions are altered as a result of drug treatment and subsequent immunomodulation. The resource is designed to aid researchers and

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clinicians in creating more effective combinatorial treatment strategies by analysing the immunomodulatory properties of individual drugs. Additionally, the database includes tools that allow users with SMILES information of a query compound to perform various types of molecular similarity searches against the SMILES data of drugs catalogued in the database.

2. METHODS

2.1 Data Collection and Compilation

To gather comprehensive, experimentally validated information on the immunomodulatory properties of anticancer drugs and antineoplastic agents, an extensive systematic methodology was implemented. Data was manually curated from publicly available databases and search engines including PubMed, Google scholar and other relevant literature sources. Predefined keyword combinations and phrases like "cancer drugs", "anti-tumour activity", "immune response", "immune checkpoint inhibitors" and "immunomodulation" were used for systematic searches. Boolean operators (NOT, AND, OR) and filters were applied to refine the search results.

The retrieved articles underwent preliminary screening, wherein titles and abstracts were reviewed to identify studies highlighting the immunomodulatory properties of drugs and antineoplastic agents. Articles that passed this initial screening were subjected to a full-text review, during which relevant data fields were identified and extracted. This process resulted in approximately 500 research articles being included for database compilation.

2.2 Database Content

To ensure clarity, accessibility and comprehensive analysis, the information in the database is systematically organised into three categories of studies: cell line studies, animal studies, and human studies. Each category is further divided into primary and secondary data. Primary data contains the information directly curated from research publications, and includes the following common data fields (i) Drug, (ii) Drug Category, (iii) Drug Bank ID, (iv) Mechanism of action (or Drug Target) , (v) KEGG ID, (vi) Pubchem ID, (vii) PMID, (viii) Dose, (ix) Technique/Assay, (x) Immune System components -Function enhanced/numbers increased (Cells, Cytokines, CD,MHC, Transcription factors etc whose function/Number/expression increased as a result of drug treatment and resulted in increased/enhanced anticancer immunity), (xi) Immune System components -Function enhanced/numbers increased (Cells, Cytokines, CD,MHC, Transcription factors etc whose function/Number/expression decreased as a result of drug treatment and resulted in decreased

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anti-cancer immunity), (xii) Immune system Unaffected (Immune components which remain unaffected on application of given treatment), and lastly (xiii) Negative immune system regulation -Function enhanced/numbers increased (Cells, Cytokines, CD,MHC, Transcription factors etc whose function/Number/expression increased as a result of drug treatment *but* resulted in decreased anti-cancer immunity). Each cell line study also includes the name of the cell line as a field. Similarly, animal studies include additional two data fields (i) Animal Strain, (ii) Cell line used to induce tumor in mice. Each patient study included additional two data fields (i) Male Female ratio, (ii) and Median age of study group.

2.3. Structural Information Mapping and Comprehensive Database Integration

To enhance the database's value, each drug entry is linked to external resources such as Pubchem, DrugBank and KEGG. PubChem provides detailed chemical information, DrugBank offers comprehensive data on drug targets and mechanisms of action, and KEGG integrates biological pathways and functional insights.

By mapping drugs to these databases, researchers gain a multi-faceted view of each drug, including its chemical structure, biological activity, mechanisms of action, and clinical relevance. This integration supports a deeper understanding and facilitates translational research, from basic science to clinical applications.

2.4 Database Architecture and Web Interface

The database is hosted on HTTP server running on a Linux operating system, ensuring stability and security for web operations. PostgresSQL [46] is used as the database management system, providing robust backend support for data storage, management and retrieval while enabling seamless data manipulation and querying. Front end of the web interface was developed using a combination of HTML, CSS and JavaScript. Python programming was employed for the backend processing and Python framework known as Django [47] was used to connect the web interface with the database.All common Gateway interface and database interface scripts were written in Python. The architecture of CancerImmunoDB is shown below in figure 2.

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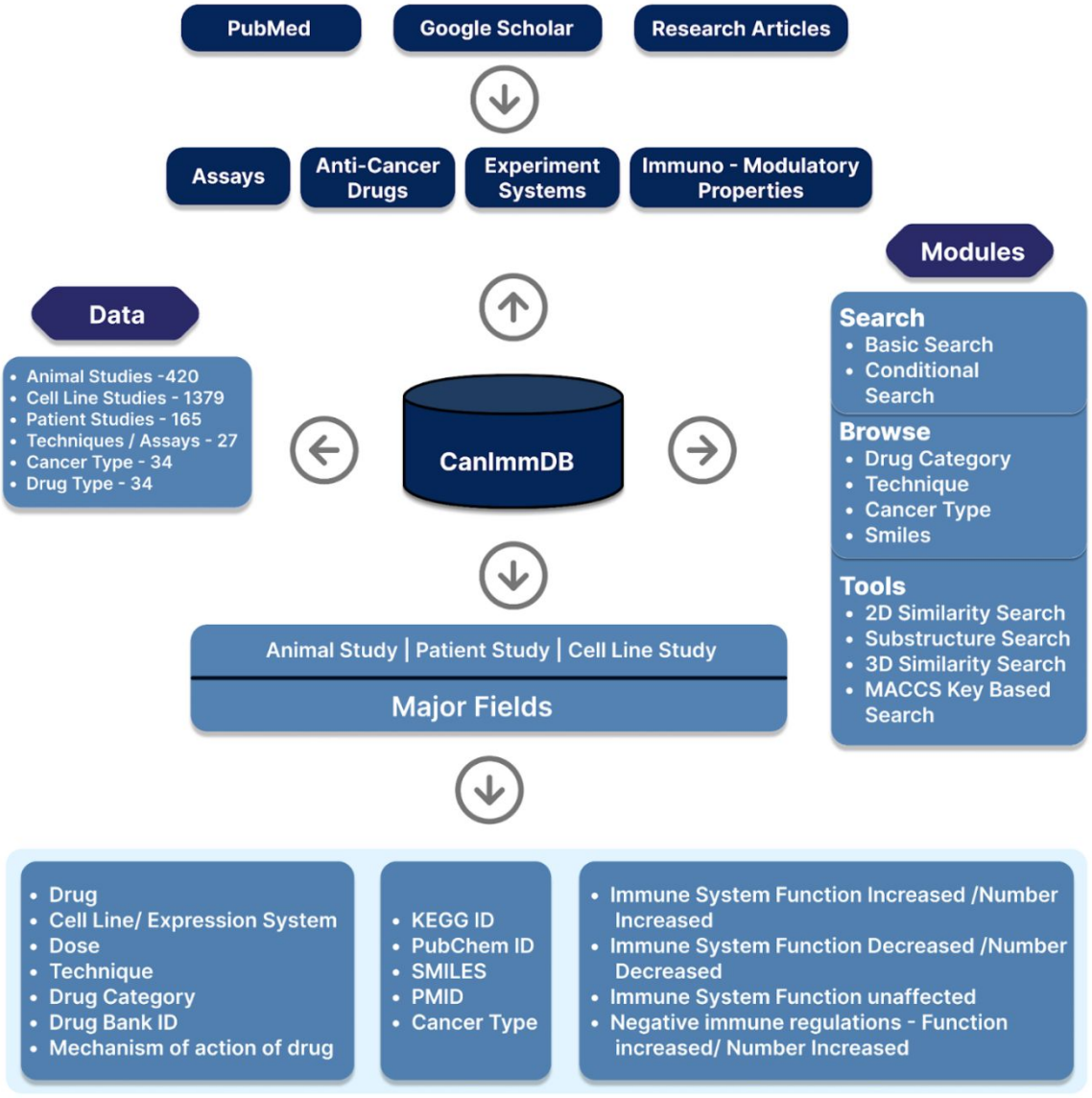


Fig 2. Architecture of CancerImmunoDB

3. RESULTS

3.1 Tool Implementation Strategy

A user-friendly web interface has been designed to query the database (Figure 3). It incorporates several tools to simplify data retrieval, searching, and analysis. The brief description of the database is given below:

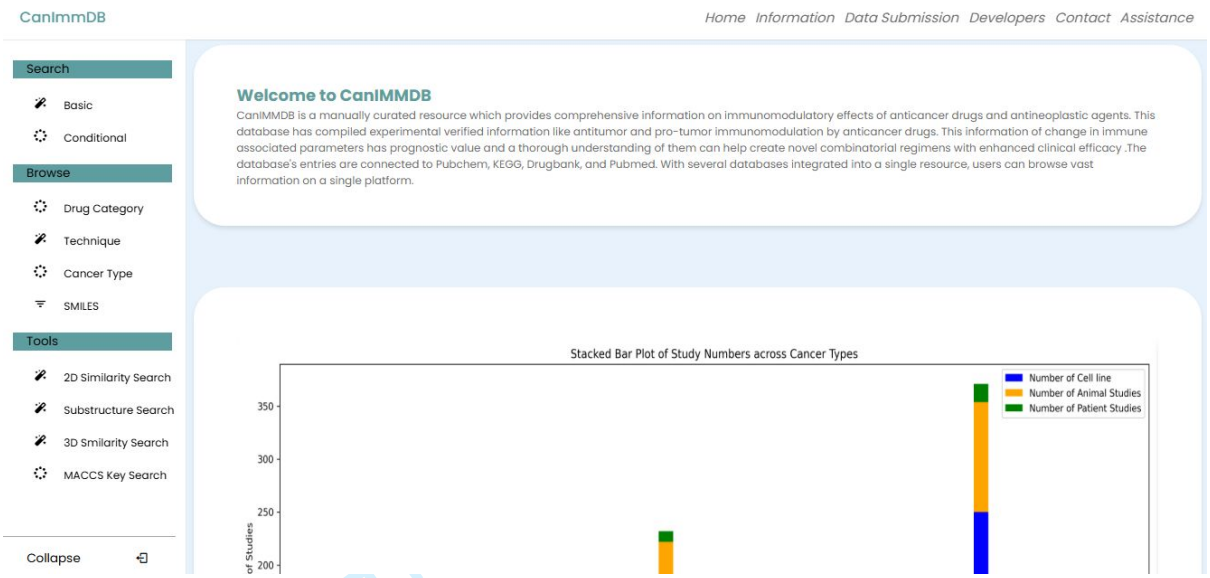


Fig 3. Overview of the CancerImmunoDB web interface.

3.2 Data access and extraction tools

3.2.1 Data Search

The **CancerImmunoDB** database includes both basic and advanced (conditional) search functionalities, ensuring efficient and user-friendly access to information. As illustrated in Figure 4, users can query the database using keywords or specific fields of interest. The **basic search module** allows users to perform straightforward searches, with results tailored to match their input query. In contrast, the **conditional search module** enables users to construct more complex queries by combining multiple search criteria using Boolean operators such as AND, OR, and NOT. This feature allows for simultaneous searches across multiple parameters, making it easier to refine and pinpoint desired data.

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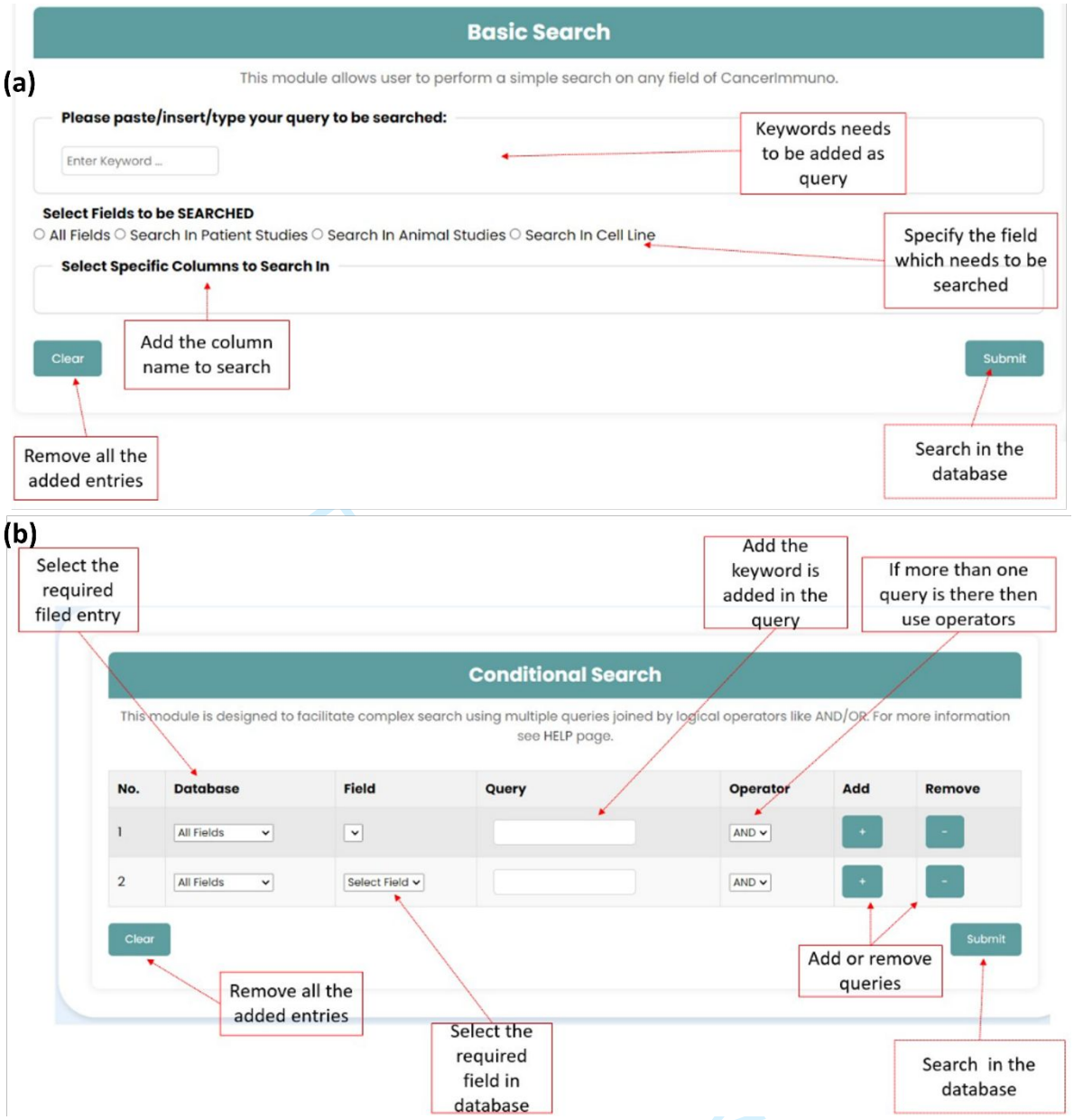


Fig 4. Data retrieval using search (a) Basic search (b) Conditional search

3.2.2 Browse

The data can be browsed by (i) Drug Category, (ii) Technique, (iii) Cancer Type, and (iv) SMILES. Each category in the CancerImmunoDB database provides specific information related to it. When browsing by SMILES, the user selects the required dataset and provides the SMILES, PubChem ID, or DrugBank ID of the compound to obtain detailed information about the entry in that dataset as shown in the figure 5.

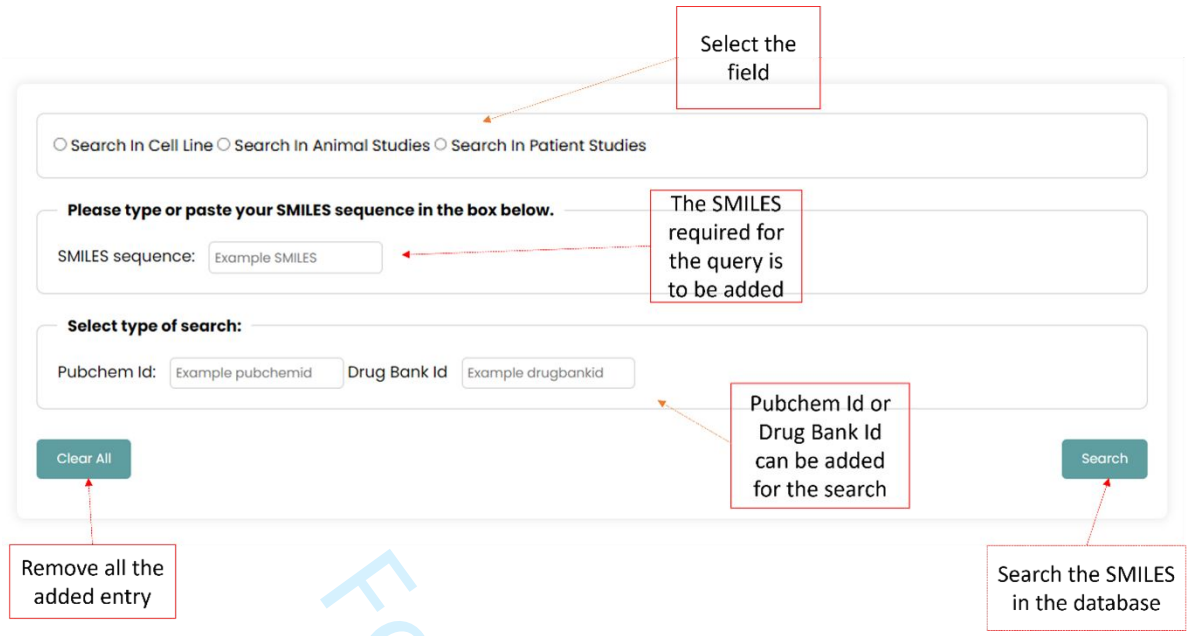


Fig 5. For the browsing with SMILES in the database.

3.2.3 Data Similarity or Substructure

The Allchem and Datastructs modules of Rdkit [48] have been implemented to facilitate similarity-based data searches. As shown in figure 6, for 2D similarity identification, threshold values must be specified, whereas no threshold is required for 3D structure searches. For substructure searches, a true or false value is provided along with the identified match. In a similar way, MACCS key search exists as well and generates MACCS keys using SMILES.

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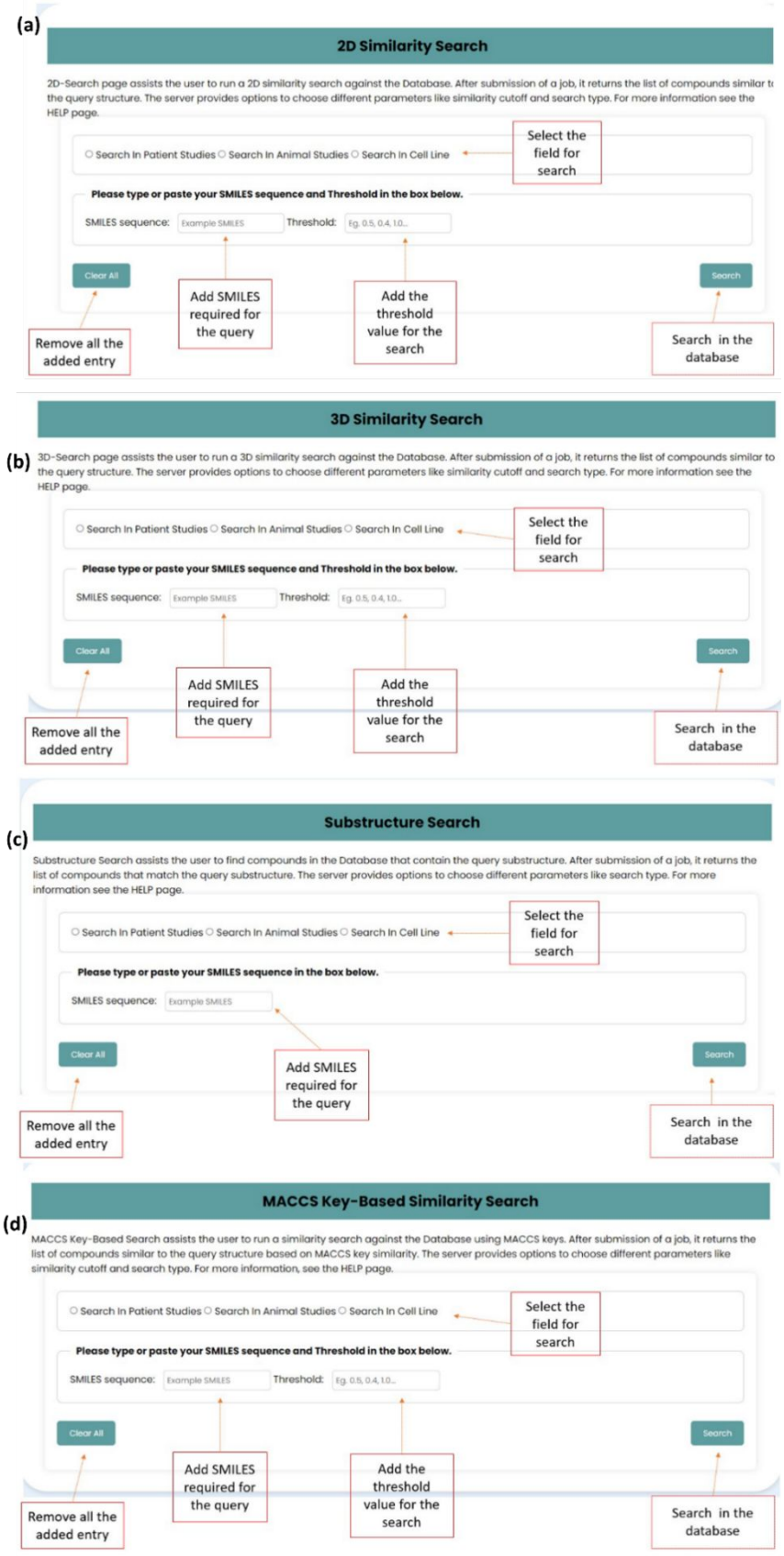


Fig 6. Tools for molecular similarity searching using SMILES (a) 2D similarity search (b) 3D similarity search (c) Substructure similarity (d) MACCS-key based similarity search

3.3 Database Statistics

In the CancerImmunoDB, the four browsing categories encompass varying numbers of studies. Within the drug categories, there are 1,221 studies involving cell lines, 406 animal studies, and 164 human studies. Similarly, under the technique category, there are 1,548 cell line studies, 432 animal studies, and 160 human studies. Furthermore, in the cancer type category, there are 1,363 studies utilizing cell lines, 411 studies involving animals, and 165 studies conducted with human subjects. Each category provides a unique perspective and scope of research, highlighting the diverse approaches and methodologies employed in cancer immunotherapy studies

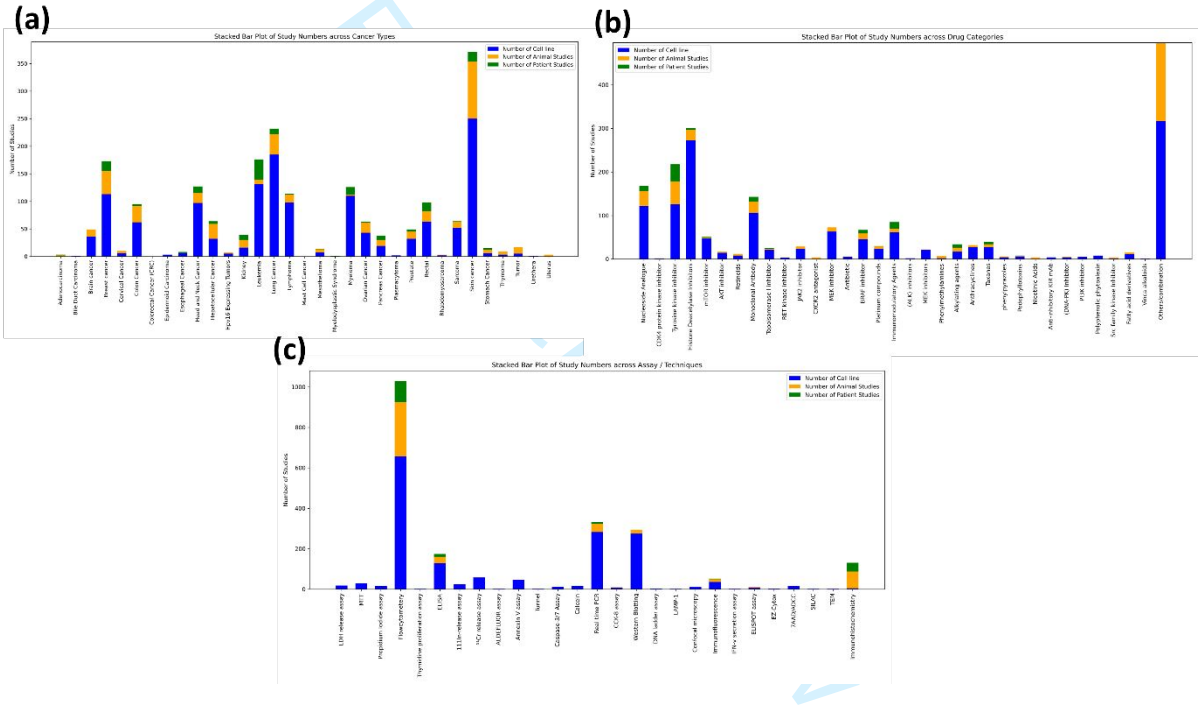


Fig 7. Distribution of Studies Across Different Browsing Categories in CancerImmunoDB (a)Cancer type (b) Category (c)Technique

3.4 Utility of Database

- First dedicated resource: To the best of our knowledge, CancerImmunDB is the first web based computational resource dedicated to studying the immune system interaction with cancer drugs and cataloguing the immunomodulatiory effects of anticancer drugs in a structured way in different systems: cell line, animal and human studies.

- Facilitating treatment plans: The structured data within the resource aids in the development of more effective combinatorial treatment strategies by enabling users to explore the immunomodulatory properties of individual drugs.
- Comprehensive Integration: The database's entries are linked to Pubchem, KEGG and Drugbank. With several databases integrated into a single resource, users can browse vast information on a single platform.
- Molecular Similarity Search: The resource offers tools for performing molecular similarity searches. Users can query compounds, based on their SMILES information and compare them with drugs listed in the database, helping identify drugs with known immunomodulatory properties.

4. DISCUSSION

Numerous databases have been developed to catalogue data on the cancer-immune interactions. However, until now, no web-based computational resource systematically indexed the scattered literature on anticancer drug immunomodulation across diverse system such as cell lines, animal models, and human studies. In this context, we introduce CancerImmunoDB, a dedicated repository of immunomodulatory drugs that fills this critical gap. CancerImmunoDB represents a significant advancement in the cataloguing and analysis of immunomodulatory effects of anticancer drugs and antineoplastic agents, offering a structured and comprehensive collection of data spanning cell lines, animal models, and human studies.

A key strength of CancerImmunoDB is its extensive coverage of different types of studies. With over 1,379 cell line studies, 420 animal studies, and 165 human studies within the drug categories alone, the database provides a rich and diverse dataset that can serve as valuable resource for researchers. The database also includes substantial data in other categories such as technique and cancer type, further enhancing its utility. This broad diversity ensures that CancerImmunoDB can accommodate a wide range of research needs, from basic mechanistic investigations to clinical applications and therapeutic strategies.

In addition to its extensive dataset, CancerImmunoDB adds substantial value by integrating detailed structural information from well-known databases like PubChem, DrugBank, and KEGG. By mapping each drug to these databases, CancerImmunoDB offers researchers access to chemical structures, biological activities, mechanisms of action, and clinical relevance. This integrated approach provides a more comprehensive understanding of how anticancer drugs

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modulate the immune system, facilitating the identification of promising therapeutic strategies and potential drug repurposing opportunities.

The database's user-friendly web interface further enhances its utility. The platform supports both basic and conditional searches, along with similarity and substructure searches, enabling users to efficiently query the data. Researchers can explore information based on drug category, technique, cancer type, or SMILES information, providing flexibility and ease of use. This functionality makes CancerImmunoDB a valuable tool for researchers at various stages of their investigations, from early drug screening to more detailed immune-response studies.

One of the most important contributions of CancerImmunoDB is its ability to highlight the prognostic value of immune-associated parameters. For example, studies have shown that the infiltration of CD16+ myeloid cells in colorectal cancer is associated with improved overall survival, while cytotoxic cell infiltration in Hodgkin's lymphoma correlates with decreased overall survival [40]. These findings underscore the potential for immune parameters to serve as predictive biomarkers for cancer prognosis. Such insights also emphasize the importance of considering immune modulation in personalized treatment strategies, aligning with the growing trend of using immune-related parameters to guide clinical decisions.

In summary, CancerImmunoDB provides a much-needed, structured, and comprehensive resource for understanding how anticancer drugs modulate the immune system across different models and populations. The integration of diverse study types, detailed structural data, and advanced search functionalities positions this database as an essential tool for advancing cancer immunotherapy research and personalized medicine.

5. CONCLUSION

CancerImmunoDB fills a crucial niche in the landscape of cancer immunotherapy research. By providing a comprehensive, curated, and easily accessible repository of immunomodulatory data, it enables researchers to explore the complex interactions between anticancer drugs and the immune system. This resource not only supports the identification of novel therapeutic targets but also facilitates the development of more effective combinatorial treatment regimens, ultimately contributing to improved clinical outcomes for cancer patients. We hope that CancerImmunoDB will be a useful resource for researchers working in the area of cancer therapeutics.

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DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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